

Variance Inflation Factor (VIF) Code Explanation

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
def calc_vif(X):
    vif = pd.DataFrame()
    vif["Variables"] = X.columns
    vif["VIF"] = [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]

    return vif
```

from statsmodels.stats.outliers_influence import variance_inflation_factor

This line imports the variance_inflation_factor function from statsmodels.

VIF (Variance Inflation Factor)

VIF helps identify:

- Multicollinearity
- High correlation between independent variables

def calc_vif(X):

This creates a function named calc_vif.

- X represents input dataset
- Usually contains independent variables

vif = pd.DataFrame()

Creates an empty DataFrame.

This DataFrame stores:

- Variable names
- Their VIF values

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```
vif["Variables"] = X.columns
```

Stores all column names from dataset into:
Variables

```
vif["VIF"] = [variance_inflation_factor(X.values, i)  
              for i in range(X.shape[1])]
```

This line calculates VIF for every column.

Important Parts

X.values

Converts dataframe into numerical array.

X.shape[1]

Represents total number of columns.

```
for i in range(X.shape[1])
```

Loops through every column one by one.

```
variance_inflation_factor(X.values, i)
```

Calculates VIF value for each variable.

```
return vif
```

Returns final VIF DataFrame output.